Fiscal Year:	FY 2018	**Task Last Updated:**	FY 06/10/2018
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Development and Testing of Biomarkers to Determine Individual Astronaut Vulnerabilities to Behavioral Health Disruptions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2014	**End Date:**	07/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Williams, Thomas	**Contact Phone:**	281-483-8773
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Flight Program:			
Flight Assignment:	NOTE: Extended to 7/31/2019 per NSSC information (Ed., 8/7/18) NOTE: Extended to 7/31/2018 per K. Ohnesorge/JSC (Ed., 7/14/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17) NOTE: End date shows as 7/31/2017 per NSSC information (Ed., 6/2/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barger, Laura Ph.D. (Brigham & Women's Hospital) Czeisler, Charles M.D., Ph.D. (Brigham and Women's Hospital/Harvard Medical Center) Duffy, Jeanne Ph.D. (Brigham & Women's Hospital) Kristal, Bruce Ph.D. (Brigham & Women's Hospital)		
Grant/Contract No.:	NNX14AK53G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	It is well established that sleep deficiency and circadian desynchrony impair alertness and performance. It is also increasingly recognized that the response to sleep loss and circadian desynchrony is highly individualized and both challenge- and task-dependent. This high degree of inter-individual variability makes it very difficult to predict who will be at risk and when they will be at risk, and therefore a 'one-size-fits-all' approach is likely to be suboptimal. What is needed are biomarkers of individual vulnerabilities and resiliencies to sleep loss and circadian desynchronization that can be used to predict problems and intervene when necessary. In this project, we will first identify candidate biomarkers in a retrospective analysis from neurocognitive and psychological measures collected in more than 1,500 subjects that have been studied over the past 17 years in our laboratory analog and from existing extensive longitudinal data in over 50 individuals overwintering in Antarctica, which is a high fidelity space analog. These, and previously untested biomarkers, will then be validated in new limited laboratory analog studies to inform a new space analog study that will test the feasibility and utility of a set of core biomarkers to predict and assess variation in neurocognitive and psychological function associated with living in Antarctica during a 12-month mission. These innovative studies have a high potential to identify reliable biomarkers that are suitable for operational use.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of a candidate biomarker panel that delineates inter-individual differences in sleep and circadian rhythms has enormous potential for the development of individualized predictive models of vulnerabilities and resiliencies to sleep deprivation and circadian misalignment for a wide range of occupational settings, for example, in shift-workers and other unusual environments (military, shipping, mining, etc).
Task Progress:	There are three main phases to our project. In Phase 1, we will identify candidate biomarkers to predict neurocognitive and psychological responses to sleep deprivation and circadian misalignment from a retrospective analysis of existing sleep, circadian, and performance data of more than 1,500 subjects that have been studied over the past 17 years. In Phase 2 of this protocol, we will identify candidate biomarkers to predict neurocognitive and psychological responses to sleep deprivation and circadian misalignment in a prospective analysis of multiple -omics data (e.g., metabolomics, lipidomics, immune response) from up to 20 subjects who will participate in a 6-day study in our laboratory analog. In Phase 3, we will assess candidate biomarkers in Antarctic expeditioners just prior to or immediately at the start of a 6-12 month overwintering period to predict individual vulnerabilities to sleep deprivation and circadian misalignment based on our findings from Phases 1 and 2. We will then collect from these individuals a number of subjective and objective measures of individual behavioral health outcomes, including sleep, circadian rhythms, urinary cortisol and melatonin levels, and a range of neurocognitive performance outcomes and alertness, to validate our predictions. In the first four years of this grant, we have completed the development of a preliminary candidate biomarker panel that can predict an individual's response to sleep loss phenotype (i.e., resilient, intermediate, or vulnerable, as defined by performance at 20 hours of wakefulness) from neurobehavioral performance collected in the individual at baseline after a habitual (8 hour) night of sleep 2 days earlier. From our analysis, we have identified a metric that can discriminate ~50% of our subject population (studied under inpatient laboratory conditions) as non-resilient to sleep loss (i.e., intermediate or vulnerable response to sleep loss) at ~20 h awake from one neurobehavioral performance assessment during rested conditions (i.e., at ~8 hours of wake after a habitual 8-hour night of sleep) with ~100% accuracy. Our findings suggest that these individuals who are poor performers during rested conditions exhibit non-sleepiness-related performance impairment that is exacerbated by subsequent sleep loss. Among the remaining ~50% of our subject population, who are high-level performers during rested conditions, the distribution of response to sleep loss phenotype is expected to be approximately equal across resilient, intermediate, and vulnerable phenotypes. A model developed to discriminate response to sleep loss phenotype at 20 h awake in these high-level performers from a single neurobehavioral performance assessment 2 days earlier during rested conditions has an overall accuracy of ~64%, suggesting that neurobehavioral performance alone is insufficient to discriminate response to sleep loss phenotype in these individuals. Our metric that discriminates high-level from poor performers during rested conditions has been implemented as a screening tool to select participants for our 6-day Phase 2 study; only high-level performers have been included as participants in our study. These high-level performers who also met all our other inclusion/exclusion criteria have been enrolled in our 6-day laboratory-based study for Phase 2 of this project. In this laboratory study, we are collecting multiple biomarker samples during baseline and sleep deprivation to determine whether there are one or more "omics"-based measures that provide a greater sensitivity and specificity to identify individuals who are resilient versus non-resilient to sleepiness-related performance impairment. We started recruiting for this study in January 2016, and have consented and screened 60 potential candidates in total. Forty-five were excluded due to not meeting study inclusion/exclusion criteria, including 8 participants who were identified as poor performers from the 35 participants who completed our initial screening test battery. To date, we have completed studies on 15 individuals. Enrollment in this study is now complete. Initial analysis of lipidomics data has identified 340 unique lipids, ~42% of which exhibit a rhythm during rested conditions and ~28% of which exhibit a rhythm during constant routine when subjects were sleep deprived. We expect to complete lipidomic and metabolomic analysis on all subjects in the next year of the project in support of Phase 2 of this study. As part of our Phase 1 analysis, we also have conducted an analysis of retrospective neurobehavioral performance data in expeditioners overwintering in Antarctica. We have developed a definition of response to sleep loss phenotype in this population based on simple reaction time performance at >16 hours of wakefulness, and developed a model to predict this phenotype from baseline neurobehavioral performance data collected 1-6 months earlier. Our current model predicts response to sleep loss with an overall accuracy of 65%. We are continuing to develop this model by incorporating information from other neurobehavioral tests at baseline to improve its overall accuracy. We plan to prospectively apply this model to baseline data collected from expeditioners who will be overwintering in Antarctica.
Bibliography Type:	Description: (Last Updated: 11/14/2025)

Abstracts for Journals and Proceedings	St. Hilaire MA, Rahman SA, Sullivan JP, Kristal BS, Quackenbush J, Duffy JF, Barger LK, Czeisler CA, Lockley SW. "Development and testing of biomarkers to determine individual astronauts' vulnerabilities to behavioral health disruptions." Presented at the 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018